

1. The following experimental items are provided for estimating the specific heat capacity of bronze c_b :

- a bronze sphere of mass 0.80 kg hung with a thread at room temperature T_0
- a polystyrene cup containing 0.50 kg of water at room temperature T_0
- a water bath maintained at 60°C
- a stirrer
- a towel

(a) Describe the procedures of the experiment and state **TWO** experimental precautions to be taken. Write down an equation for finding c_b .
Given: specific heat capacity of water = $4200 \text{ J kg}^{-1}^\circ\text{C}^{-1}$ (5 marks)

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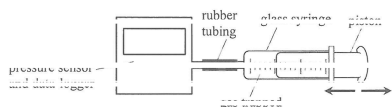
(b) The value of c_b found in the experiment in (a) is lower than the actual value. Explain. (2 marks)

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Figure 2.1



Judy uses the set-up shown in Figure 2.1 to study the relationship between the pressure and volume of a fixed mass of gas at constant temperature. The volume V of the gas trapped is read directly from the syringe and the corresponding pressure p is measured by a data-logger via a pressure sensor.

- (a) The initial volume and pressure of the gas are $6.0 \times 10^{-5} \text{ m}^3$ and $1.0 \times 10^5 \text{ Pa}$ respectively at a room temperature of 25°C . Estimate the number of gas molecules trapped in the syringe. (3 marks)

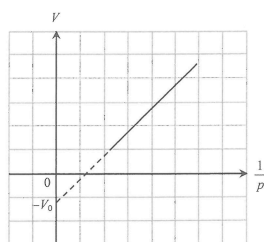
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- (b) The piston is then pushed in or pulled out to vary V and p such that several pairs of readings are recorded.

Figure 2.2 shows the graph of V against $\frac{1}{p}$ plotted.

Figure 2.2



- (i) State ONE experimental precaution for keeping the gas temperature constant. (1 mark)

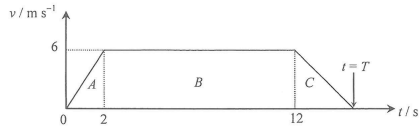
- (ii) The straight line graph does not pass through the origin but cuts the vertical axis at $-V_0$ instead. Suggest what V_0 stands for. (1 mark)

- (iii) If the experiment is repeated at a higher room temperature using this set-up with the same mass of the same gas, sketch the expected graph in Figure 2.2. (2 marks)

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3. A person of mass m stands on a balance inside a lift. The lift goes down from the top of a building at time $t = 0$ and it reaches the ground at $t = T$. The velocity-time (v - t) graph of the lift is shown in Figure 3.1. ($g = 9.81 \text{ m s}^{-2}$)

Figure 3.1



- (a) Calculate the acceleration of the lift from $t = 0$ to $t = 2 \text{ s}$. (2 marks)

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The reading of the balance changes during the person's ride on the lift and the readings registered are 685 N, 569 N and 795 N.

- (b) Match these readings with the three stages, A, B and C, of the ride (shown in Figure 3.1). Hence deduce the mass of the person. (3 marks)

A: _____ B: _____ C: _____

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- (c) (i) Show that $T = 15 \text{ s}$. (2 marks)

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- (ii) Hence estimate the height of the building. (2 marks)

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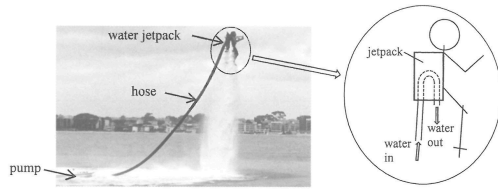
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Figure 4.1



A person wears a water jetpack which enables him to stay 'afloat' in equilibrium in the air as shown in Figure 4.1. A pump on the sea surface continuously pumps water to the jetpack via a hose and the water is

- (a) Referring to Figure 4.1, water enters the U-shape hose inside the jetpack with a certain speed and is then ejected out vertically downwards. Use Newton's law(s) of motion to explain why a lifting force acting on the person is produced. (2 marks)

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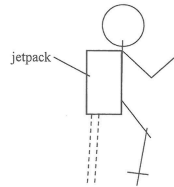
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- (b) Draw and label all the forces acting on the person wearing the jetpack as a whole in the free-body diagram below. Neglect the pulling force due to the hose connected to the jetpack. (1 mark)



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- (c) Suppose that water enters the jetpack with a speed of 10 m s^{-1} vertically upwards and is then ejected out at the same speed vertically downwards. ($g = 9.81 \text{ m s}^{-2}$)

- (i) Just by considering the change of momentum of the water, estimate how much water, in kg, has to be ejected per second to provide a lifting force of 1000 N needed. (2 marks)

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- (ii) Water is pumped to the water jetpack at a height of 7.5 m above sea surface and then ejected from it. By considering the gain in mechanical energy of the water, estimate the minimum output power of the pump. (2 marks)

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- (d) The person changes to staying 'afloat' in equilibrium at a higher position. If the speed by which water enters and is ejected from the jetpack remains the same, would the amount of water ejected per second be greater than, equal to or smaller than the result found in (c)(i)? Explain. (Neglect the weight of the hose.) (2 marks)

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Figure 5.1

Kitty uses the set-up in Figure 5.1 to study the image formation of a lens. The lens is placed at a distance u from an illuminated object (letter 'P'). An opaque screen is placed at a distance D from the object so as to capture the image.

(a) (i) State the kind of lens used. Explain your answer. (2 marks)

(ii) Sketch the image on the screen seen by the observer. (1 mark)

(b) The separation D is varied while the position of the lens is adjusted to form an image once again on the screen. The corresponding object distance u is obtained for plotting a graph of D against u (Figure 5.2).

Figure 5.2

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(i) When the lens is placed at 18 cm from the object, use the graph to find the corresponding separation between the lens and the screen. Hence calculate the magnification of the image. (2 marks)

In the figure below, AB represents the illuminated object which is at 18 cm from the lens L . p , q and r are light rays.

(ii) Indicate the image formed by AB (denote it as I) and draw the refracted rays of p , q and r . (3 marks)

(iii) Hence find the focal length of the lens. The horizontal scale is 1 cm to 5 cm. (1 mark)

Focal length = _____

(iv) Keeping the object and screen fixed in position, suggest where Kitty should move the lens to such that an image can be formed again on the screen. State the ratio $\frac{\text{height of this new image}}{\text{height of the original image}}$. (2 marks)

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- *6. (a) A laser beam is directed perpendicularly towards a double slit of separation $a = 0.3 \text{ mm}$. The pattern of bright spots projected on a screen 1.8 m away from the slits is shown in Figure 6.1.

Figure 6.1



- (i) Find the wavelength of the laser beam. (3 marks)

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- (ii) Explain why the slit width has to be very narrow in order for the above pattern to be observed. (2 marks)

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- (b) The double slit is now replaced by a diffraction grating of 500 lines per mm .

- (i) Find the separation between the central bright spot and first-order bright spot of the pattern on the screen for the same experimental settings. (3 marks)

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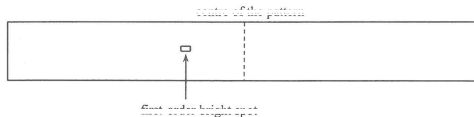
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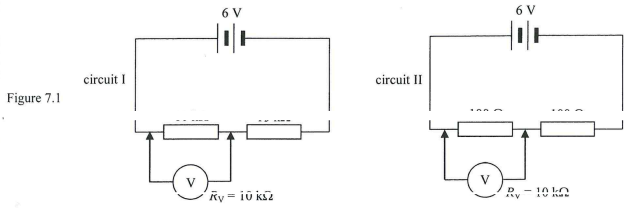
- (ii) Sketch the pattern, up to the second-order, that you would expect to see on the screen when using this diffraction grating. A first-order bright spot has already been drawn for you. (2 marks)



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7. (a) The circuits in Figure 7.1 each contains two resistors connected in series with a 6 V battery of negligible internal resistance. The resistors in circuit I are $10\text{ k}\Omega$ each while those in circuit II are $100\text{ }\Omega$ each.



A voltmeter of internal resistance $R_V = 10\text{ k}\Omega$ is used to measure the potential difference across one of the resistors as shown.

- (i) What would be the respective voltmeter readings? (3 marks)

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- (ii) In fact, the potential difference across each resistor *before connecting the voltmeter* is 3 V in both circuits. Explain why this voltmeter gives a relatively inaccurate value for circuit I. Hence state the general principle of selecting a suitable voltmeter for such measurement. (2 marks)

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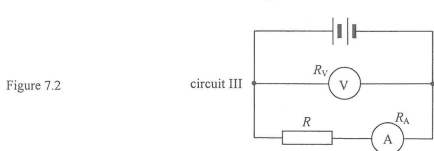
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- (b) Circuit III shows a possible method for measuring resistance using a voltmeter and an ammeter. The internal resistances of the voltmeter and the ammeter are R_V and R_A respectively and their readings V_m and I_m give the measured resistance $R_m = \frac{V_m}{I_m}$. The true resistance value of the resistor is R .



- (i) State which reading(s), V_m , I_m or both, do(es) **NOT** give the *true voltage* across the resistor and/or the *true current* passing through the resistor. Hence write down an equation relating R_A , R_m and R . (2 marks)

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- (ii) Find the percentage error associated with R_m when measuring the resistance of this resistor. Given: $R_V = 10\text{ k}\Omega$, $R_A = 1\text{ }\Omega$ and $R = 10\text{ }\Omega$. (2 marks)

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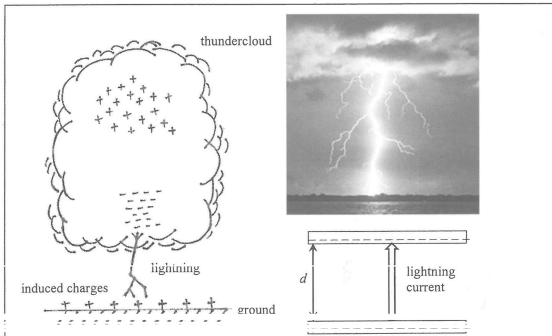
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8. Read the following passage about lightning and answer the questions that follow.



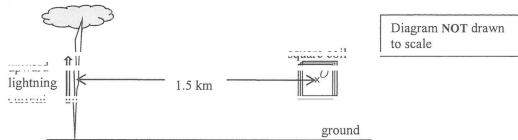
Lightning occurs when charges accumulate in the clouds to such an extent that the electric field in the atmosphere is strong enough to cause the air to lose its insulating properties. The threshold electric field for 'breakdown' to occur is about $3 \times 10^5 \text{ V m}^{-1}$ above which electrons or ions in the atmosphere can pass through the air between clouds and the ground or between clouds and clouds. The peak current of a typical lightning bolt can reach about 30000 A. How the charges are separated and accumulated in the clouds is not fully understood yet. In most cases, negative charges are at the base of the cloud and positive charges are induced on the ground.

- (a) (i) What is the meaning of 'breakdown' in the passage? (1 mark)

* (ii) The thundercloud's base and the ground can be modeled as two parallel plates with opposite charges. If the negative charges distributed at the cloud's base are about $d = 2 \text{ km}$ from the ground, find the potential difference between the cloud and the ground when the electric field in the atmosphere just reaches the threshold of 'breakdown'. (2 marks)

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A lightning detector having a small square coil inside is placed at point O which is 1.5 km from the lightning bolt. The coil and the lightning's direction are in the same vertical plane as shown. Assume that the lightning current flows vertically upwards to the thundercloud from the ground.



- (b) (i) State the direction of the magnetic field (to the left / to the right / into paper / out of paper) produced at point O by the lightning current. Estimate the magnetic field strength's peak value at O . (3 marks)

(ii) Explain why within the very short duration of lightning an induced current first flows in the coil in a certain direction and then reverses. Your answer should include the directions of the induced current in the coil. (2 marks)

(iii) Among the physical quantities related to lightning, electric field in the atmosphere, lightning current and magnetic field due to lightning, suggest which one can be monitored so as to give fore-warning of lightning. Explain your choice. (2 marks)

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- Figure 9.1



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